

More Agents, Less Accuracy: When Multi-Agent LLM Voting Hurts on Knowledge-Bound Tasks

COMMON PRACTICE

"More Agents Is All You Need" — Li et al., 2024



↑ **more accuracy**
(claimed)

Monotone gains reported on broad benchmarks $\Rightarrow$ default architectural recipe

BUT

OUR FINDING

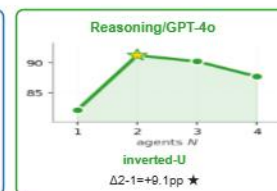
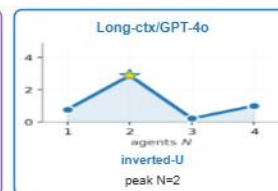
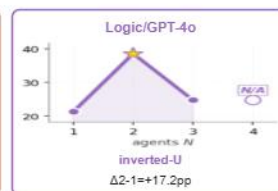
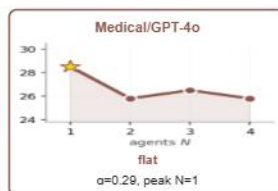
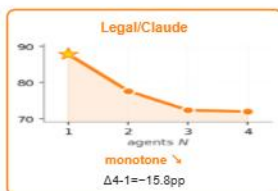
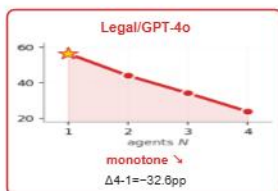
On knowledge-bound NLP, voting hurts accuracy



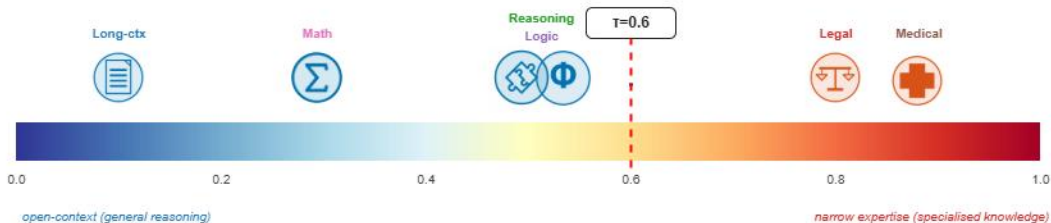
↓ **LESS accuracy**
-32.6 pp

on legal/GPT-4o ($N=1 \rightarrow 4$, McNemar $p < 10^{-100}$)

Evidence: same protocol, different shapes — six (domain $\times$ model) cells



Recipe: predict N^* from a single text-judged feature K in ≈ 15 min



Decision rule

$K < 0.6$
 $N^* = 2$
voting helps

$K \geq 0.6$
 $N^* = 1$
extra agents harm



Verify with paired pilot: $r > \alpha(1-p) / (1-\alpha)$

$\Delta = (1-\alpha)r - \alpha(1-p)$

K scored from task description by two LLM judges (inter-judge $r=0.885$).

7/8 main cells (LOO); 9/12 combined w/ tier-2